$$0.3667 \cdot \varphi_2 - 0.2 \cdot \varphi_3 - 0.1 \cdot \varphi_4 = 2.6667$$
$$-0.2 \cdot \varphi_2 + 0.4778 \cdot \varphi_3 - 0.1667 \cdot \varphi_4 = -18.2222$$
$$-0.1 \cdot \varphi_2 - 0.1667 \cdot \varphi_3 + 0.3167 \cdot \varphi_4 = 16.6667$$

Где

$$G_3 = \frac{1}{R_3} = \frac{1}{15} = 0.066667\,S$$

$$G_2 = \frac{1}{R_2} = \frac{1}{9} = 0.111111\,S$$

$$G_5 = \frac{1}{R_5} = \frac{1}{20} = 0.05\,S$$

$$G_1 = \frac{1}{R_1} = \frac{1}{6} = 0.166667\,S$$

$$G_6 = \frac{1}{R_6} = \frac{1}{10} = 0.1\,S$$

Решим систему с помощью матриц

$$\varphi = (A G_B A^T)^{-1}(-A J_B - A G_B E_B)$$

После расчета получим

$$\varphi_2 = 10.551\,V$$

$$\varphi_3 = -17.394\,V$$

$$\varphi_4 = 46.809\,V$$

Определяем токи ветвей

$$I_3 = \frac{\varphi_1 - \varphi_2 + E_3}{R_3} = \frac{0 - (10.551) + 100}{15} = 5.963\,A$$

$$I_2 = \frac{\varphi_3 - \varphi_1 + E_2}{R_2} = \frac{-17.394 - (0) + 50}{9} = 3.623\,A$$

$$I_5 = \frac{\varphi_4 - \varphi_1}{R_5} = \frac{46.809 - (0)}{20} = 2.34\,A$$

$$I_1 = \frac{\varphi_3 - \varphi_4 + E_1}{R_1} = \frac{-17.394 - (46.809) + 100}{6} = 5.966\,A$$

$$I_6 = \frac{\varphi_4 - \varphi_2}{R_6} = \frac{46.809 - (10.551)}{10} = 3.626\,A$$

$$I_0 = (\varphi_3 - \varphi_2) G_0 = (-17.394 - (10.551)) \cdot 0.2 = -5.589\,A$$

$$J_0 = J_0 = 4\,A$$